

Training data The MACE-MP-0b3 model was trained on the MPtrj dataset which was compiled originally for CHGNet (95). This dataset consists of a large number of static calculations and structural optimization trajectories from the Materials Project (MP) (93). These include approx. 1.5M configurations (roughly ten times the approx. 150k unique MP structures), mainly small periodic unit cells (90% under 70 atoms) describing inorganic crystals with some molecular components. The DFT calculations use the PBE exchange-correlation functional with Hubbard U terms applied to some transition metal oxide systems, but no additional dispersion correction (144).

Since the potential we fit calculates the energy based only on structural information, ideally we would like to use consistent electronic calculation parameters and the lowest energy electronic state for each configuration. One significant source of inconsistency is the application of Hubbard U , which is used in MP calculations only when O or F are present together with any of 8 transition metals (Co, Cr, Fe, Mn, Mo, Ni, V, W) (145). The application of U leads to a shift in energy correlated with the value of U , *i.e.* a few eV, not explicitly accounted for in our fit. Thus, energies from calculations using those 8 elements with and without O or F are inconsistent (in the sense that the energy along a continuous deformation path that removes the O or F atoms from around these metals would be discontinuous). The pre-trained CHGNet fit to MPtrj used energies corrected to account for the presence or absence of U (146). In our fit, this shift only occurs between structures with different compositions and for any given composition the energies should be consistent. As a result, we expect configurations that include local regions of these metals with very different O or F content, *e.g.* an interface between a metal and an oxide, may be poorly described.

In addition, the current fitting database includes a variety of magnetic orders generated as part of a systematic search for the magnetic ground state (147), chosen from the full database only based on calculation type (“GGA Static” and “GGA Structure Optimization”) and energy-difference criteria (95). To quantify the effect of this additional and unaccounted-for degree of freedom, we classify the magnetic order associated with each calculation task into one of four categories: 1) no atomic magnetic moment listed, 2) moment converged to zero on all atoms, 3) converged to ferromagnetic order, and 4) converged to another magnetic order. Of the approx. 150k MP-IDs present, about 48k have more than one magnetic order present in the fitting database. In the vast majority of cases, this includes a calculation where the moments are *unknown* (*i.e.* not recorded) and a single other magnetic order, and we can hope that they are actually consistent. However, for 5186 MP-IDs we find multiple non-trivial magnetic orders. To quantify the effect on the fitting quantities, we calculate the minimum energies of each magnetic order for each material, and analyze the range of minima values seen for each material (distribution is plotted in SI Fig. S69). While the vast majority of materials have negligible variation, there are hundreds with variation >100 meV/atom (*i.e.* an order of magnitude larger than the energy error on the validation set), and a few that vary by <0.5 eV/atom.

Long-range dispersion corrections Dispersion interactions, sometimes called van der Waals interactions, are crucial for describing the weak, long-range interactions between electrons. Common approximations in DFT, such as PBE (118), cannot capture such long-ranged interactions, motivating the use of additive non-local corrections, such as DFT-D3 (29) or rVV10 (148). Inclusion of a dispersion correction to DFT is necessary to describe the dynamics of liquid water (149), the geometries and binding energies of layered solids (150), and stability of metal-organic frameworks (151), among many other examples.

Additive dispersion corrections typically employ a physical model for dispersion interactions with empirical parameters optimized to cut off the correction at interatomic distances where approximate DFT is reliable. DFT-D3 is an interatomic potential which uses tabulated values of atomic polarizabilities to describe two-body and, optionally, three-body Axilrod-Teller (152) dispersion interactions. As MACE-MP-0b3 is trained to PBE energies, forces, and stresses, it inherits PBE’s lack of long-range dispersion interactions. An optional, additive DFT-D3 dispersion correction can be applied to MACE-MP-0b3. The PyTorch implementation of DFT-D3 used in this work is described in Ref. (102). The same parameters used in PBE-D3(BJ), *i.e.*, DFT-D3 with a Becke-Johnson damping function (153), are used in the D3 correction to MACE-MP-0b3.

Author contributions

Model training: YC, PB, IB, CL; **Data/Model analysis:** PB, YC, JR, NB, RE, MCK, ES, IB; **MACE code:** IB, YC, SWN, DPK, PB, WCW, MA, SV, ES, CL; **Application examples:** WJB (CsPbI₃, appendix A.5); LLS (catalysis: In₂O₃, section 2.2 and appendix A.24.4); IB (a-C quenches, appendix A.3.1); ZEM (a-C graphitisation, appendix A.3.2); NK (Si interstitials, appendix A.1); EVU, XRA, NON (aqueous interfaces, section 2.1 and appendix A.19); YC (molten salts, appendix A.20); CSc, VK, FDP, XRA (water and ice, section 2.1 and appendix A.16); SPN and AGS (LiNiO₂, appendix A.12); SWN (lithiated graphite, appendix A.11); AME (zeolites appendix A.10); JJ (transition metal dichalcogenides, appendix A.26); JHM (ethanol/water, appendix A.17, trialanine, appendix A.30); GL, LAMR (a-Si appendix A.2); DK (carborane, appendix A.25, ammonia-borane, appendix A.23); VC (S polymerisation, appendix A.9); JR, JaG, JoG and AAN (phonons, appendix B.1); JR, REAG (materials discovery: formation energy, appendix A.29.1); KSJ (materials discovery: stoichiometric substitutions, appendix A.29.2); ADK (materials discovery: highly-coordinated structures, appendix A.29.3); ASR, YC and AME (MOFs, section 2.3 and appendix A.28); MV (solvent mixtures, appendix A.18); DPK, ES, SMB (hydrogen combustion, appendix A.8); NKa, CSu (HOIPs, appendix A.6); FF, JT (protein dynamics and stability appendix A.7); PG, YW, TDS, JRK and CO (point and extended defects in BCC metals, appendix A.13); BXS, FB (molecule-surface interactions, appendix A.31); WCW (HEA, appendix A.32); EF, SD (catalysis: linear scaling relationships, section 2.2 and appendix A.24.2); HJ, HHH (catalysis: CO oxidation on Cu, section 2.2 and appendix A.24.3); SH, SD (catalysis: Pourbaix diagrams, section 2.2 and appendix A.24.1); MCK (benchmarks: bulk and shear moduli, appendix B.2); FDP (benchmarks: cohesive energies and lattice constants of solids, appendix B.3, atomization energies appendix B.4, and reaction barrier heights, appendix B.5); TKS (Al₂O₃, appendix A.14, diatomics, appendix B.6); JPD (Arsenic random structure search, appendix A.15); IBM (high-pressure hydrogen, appendix A.22); IBM, CvdO (electrode-electrolyte interface / battery system, appendix A.27); JK, VS and KH (CeO₂, appendix A.4); FZ (ionic liquids, appendix A.21); **Fine-tuning:** IB, NB, TW **Uncertainty quantification:** FBi, MC, SC, CHH, CO and YW (appendix D); **Supervision of research:** AB, ACF, AM, ASR, CH, CO, CPG, CSu, GC, HHH, JaG, JK, JTM, KAP, KH, KR, MA, MC, SD, SMB, TV, VLD, WH; **Drafted manuscript:** IB, NB, YC, GC, SD, HHH, MCK, JR, ASR, CSc, JTM; **Edited manuscript:** IB, NB, YC, GC, VLD, JaG, REAG, JR, MCK, KAP, ASR, LLS, JTM, AM, CO, AME, WCW, JLAG; **Supervised manuscript writing:** NB, GC, VLD, VK, JTM, CSc.

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